

The Irreducible Boundary of Bounded Phase Space: Division, Boundary Thickness, and the Resolution Floor

Kundai Farai Sachikonye
Technical University of Munich
Chair of Proteomics and Bioanalytics
kundai.sachikonye@wzw.tum.de

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Abstract

We prove that to know a thing is to divide it from everything else, that every such division requires a boundary of strictly positive thickness, and that the contents of that thickness form an irreducible residue which no bounded knower can eliminate. The development is topological and measure-theoretic and rests on a single hypothesis—that the relevant state space is bounded and finitely resolvable (the *bounded phase space* hypothesis). From it we derive a *Boundary-Thickness Theorem*: in a bounded space carrying a finite resolution, no two complementary regions can be separated by a set of zero content; every separator carries positive content $\beta > 0$, the *resolution floor*. We give three independent derivations of the floor—a representational one (faithful images of a consistent invariant cannot be the invariant), a thermodynamic one (the cost of compressing a separator to zero content diverges, and a single binary distinction costs at least $k_B T \ln 2$), and a cardinality one—and prove they bound the *same* constant. We prove a *Detectability Theorem*: a bounded object registers (“is distinguishable”) if and only if its scale exceeds its own boundary thickness; sub-floor objects are engulfed by their separator and cannot be individuated. The classical sorites paradox is then a corollary, not a puzzle: there is no sharp grain at which a heap appears because the heap–non-heap separator has positive thickness, and detection is emergence from that thickness. We prove an *Inquiry-Cessation Theorem*: thinning a separator toward zero content has diverging cost and bounded return, so a finite knower rationally halts at the floor, whence knowledge attaches to bounded wholes and never to sub-floor constituents—a structure we show is visible in ordinary attribution. Finally we exhibit the physical realisation: the same hypothesis forces oscillatory dynamics, a discrete partition spectrum, the coordinate quadruple (n, ℓ, m, s) with shell capacity $2n^2$, an exclusion principle, the compression law, and a Fisher embedding—so that the floor of the epistemic theory and the floor of the physical theory are one constant. Every claim is proved from stated definitions; uncertain material has been omitted.

Keywords: bounded phase space; topological boundary; boundary thickness; resolution floor; sorites paradox; vagueness; partition; compression cost; Landauer bound; information geometry; irreducible residue.

Contents

1 Introduction

1.1 Thesis

This paper defends, and proves under a single explicit hypothesis, the following chain.

- (i) *To know a thing is to divide it from everything else.* Knowledge of an object A is the capacity to tell A from not- A ; this is a partition of the state space into A and its complement.
- (ii) *Every division requires a boundary, and the boundary has positive thickness.* In a bounded, finitely resolvable space there is no zero-content separator between a region and its complement. The separator carries a strictly positive content $\beta > 0$.
- (iii) *The boundary's interior is an irreducible residue.* The contents of the thickness belong determinately to neither A nor not- A . They are the operational form of the limitative “residue” familiar from logic and physics.
- (iv) *Detection is emergence from the boundary.* A bounded object is distinguishable exactly when its own scale exceeds its boundary thickness. Below that scale it is engulfed by its separator and cannot be individuated.
- (v) *Inquiry rationally halts at the floor.* Thinning a separator toward zero content costs without bound for bounded return; a finite knower stops, and knowledge thereafter attaches to bounded wholes, never to their sub-floor parts.

The single hypothesis is that the space in question is bounded and admits a finite-resolution partition—the *bounded phase space* hypothesis (Axiom 1). Everything else is theorem.

1.2 Why “thickness”

The everyday and the technical meet at one point. A single grain of wheat makes no sound when dropped; a heap makes a sound. Classically this is a paradox: adding one grain never converts silence to sound, yet a heap sounds. We resolve it by inverting the usual reading. The grain is silent not because it is small in the absolute, but because its *boundary is too thick relative to it*: the layered separator one would have to resolve to isolate “this grain” from “everything else” is larger than the grain, so the grain is buried inside its own boundary and never stands out as a figure against a ground. The heap sounds because it is large compared with the same separator: it clears its boundary and registers. There is no critical grain because “registering” is emergence across a band of positive width, not a crossing of a point. The width of that band is the floor β . This is ?? below, and the sorites paradox is its ??.

1.3 Why “bounded”

Boundedness is not a convenience; it is the source of the floor. On an unbounded space one could, in principle, separate a region from its complement by an arbitrarily thin frontier and resolve structure to arbitrary depth at fixed cost. It is precisely the finiteness of the accessible space—and the finiteness of the resolution any embedded knower can purchase—that forbids the zero-thickness cut. We make the hypothesis exactly strong enough to force the floor and no stronger, and we mark, at each step, where boundedness is used.

1.4 Two faces of one constant

The floor β has an epistemic face and a physical face, and a central aim of the paper is to prove they coincide. Epistemically, β is the smallest distinction a bounded knower can draw—the thickness of its sharpest boundary. Physically, the same hypothesis (Axiom 1) forces a partition spectrum, a compression law whose cost diverges as a cell is squeezed to zero measure (??), and a minimum energy $k_B T \ln 2$ per binary distinction (??); these pin a physical floor. We prove (??) that the epistemic and physical floors are not two constants but bounds on one.

1.5 Relation to existing treatments

The sorites paradox and vagueness have large literatures: degree theories and fuzzy logic (?), supervaluationism (??), epistemicism (?), and the classical analyses of ?; see ? for a survey. Our account is none of these: it is not a logic of partial truth, nor a claim that a sharp boundary exists but is unknowable, but a *geometric* statement that the separator is a set of positive content whose interior is determinately category-less. The residue connects structurally to limitative results in logic (??) and to the thermodynamics of computation (??); we use these as analogues and, in the physical sections, as tools, never as premises. The topological and measure-theoretic machinery is standard (?????), as is the ergodic and spectral background (?????) and the information geometry (???).

1.6 Method and discipline

The paper is written to a single standard: every substantive statement is a definition, a theorem proved here, or a citation of a standard result compared against such a theorem. Where a tempting claim could not be proved at this standard it has been removed rather than hedged. Physical interpretation is confined to remarks and to ??; the mathematical spine (Sections 2 and 3 and ????????) stands without it.

1.7 Organisation

Section 2 fixes the topological and measure-theoretic setting and states the bounded phase space hypothesis. Section 3 proves the Boundary–Thickness Theorem. ?? gives the three derivations of the floor and proves their agreement. ?? proves the Detectability Theorem and resolves the sorites. ?? develops the topology of the residue (the boundary interior) and its non-attribution. ?? proves the Inquiry–Cessation Theorem and the bounded-attribution capstone. ?? exhibits the physical realisation. ?? states scope and limitations and concludes.

2 The setting: bounded phase space and finite resolution

We work throughout in a metric-measure space. The objects of the theory are regions (subsets), their separators (boundaries), and partitions; the single hypothesis constrains the ambient space to be bounded and finitely resolvable.

2.1 Ambient space

Definition 2.1 (Phase space). A *phase space* is a triple (Ω, d, μ) where (Ω, d) is a metric space and μ is a Borel measure on Ω that is positive on non-empty open sets and finite on bounded sets. We write $\mathcal{L}$ for Lebesgue measure when $\Omega \subseteq \mathbb{R}^m$, and $\mathcal{H}^s$ for s -dimensional Hausdorff measure (??).

Definition 2.2 (Region, complement, separator). A *region* is a Borel set $A \subseteq \Omega$ with $\mu(A) > 0$ and $\mu(\Omega \setminus A) > 0$ (so both A and its complement are non-negligible). Its *topological boundary* is $\partial A = \text{cl}(A) \cap \text{cl}(\Omega \setminus A)$. A *separator* of A is any Borel set Σ with $\Omega \setminus \Sigma = U_A \sqcup U_{A^c}$ a disjoint union of two non-empty relatively open sets with $A \setminus \Sigma \subseteq U_A$ and $(\Omega \setminus A) \setminus \Sigma \subseteq U_{A^c}$. A separator is *sharp* if $\mu(\Sigma) = 0$.

Remark 2.3. ∂A is the smallest closed separator in the topological sense; a general separator may be thicker. The question the paper turns on is whether a *sharp* separator (one of zero measure) can exist for the regions that an embedded knower actually uses. The answer, under Axiom 1, is no.

2.2 Partitions and resolution

Definition 2.4 (Finite measurable partition). A *finite measurable partition* of Ω is a finite family $\mathcal{P} = \{c_1, \dots, c_N\}$ of pairwise-disjoint Borel sets with $\bigcup_i c_i = \Omega$ and $\mu(c_i) > 0$ for each i . A partition $\mathcal{P}'$ *refines* $\mathcal{P}$, written $\mathcal{P}' \preceq \mathcal{P}$, if every cell of $\mathcal{P}'$ is contained in a cell of $\mathcal{P}$.

Definition 2.5 (Resolution). A phase space *carries resolution* $\delta > 0$ for a partition $\mathcal{P}$ if every cell $c \in \mathcal{P}$ satisfies $\text{diam}(c) \geq \delta$; equivalently, no two points closer than δ are forced into distinct cells. The resolution is the finest scale at which the partition distinguishes points.

Remark 2.6 (Resolution is a knower-relative datum). Resolution is not a property of Ω alone but of the partition a knower can realise. Two knowers of the same space may carry different δ . The floor we derive will inherit this relativity: its magnitude depends on δ , but its strict positivity will not.

2.3 The hypothesis

Axiom 1 (Bounded phase space). The phase space (Ω, d, μ) satisfies:

- (BPS1) **Boundedness.** Ω is bounded and $0 < \mu(\Omega) < \infty$; moreover $\text{cl}(\Omega)$ is compact.
- (BPS2) **Finite resolution.** Every partition realisable by an embedded knower carries some resolution $\delta > 0$; there is no partition into cells of arbitrarily small diameter at fixed, finite descriptive cost.
- (BPS3) **Hierarchical refinability.** There is a sequence of finite measurable partitions $\mathcal{P}_0 \succeq \mathcal{P}_1 \succeq \mathcal{P}_2 \succeq \dots$, each refining the last, generating the Borel σ -algebra modulo μ -null sets; each $\mathcal{P}_n$ carries a resolution $\delta_n > 0$.

Remark 2.7 (Status of the hypothesis). (BPS1) is the substantive geometric content: the accessible space is finite in extent and measure, and its closure is compact (so the extreme-value and finite-subcover arguments below apply). (BPS2) is the finitude of any embedded knower: a knower with finite descriptive resources cannot specify cells of vanishing diameter—this is what forbids the limiting zero-thickness cut. (BPS3) states that finer partitions exist and exhaust the measurable structure, but each *individual* partition is still finite-resolution; refinement is unbounded in the limit but bounded at every finite stage. The three clauses are exactly what is needed for the Boundary–Thickness Theorem and nothing more.

Lemma 2.8 (Finiteness of cell count). *Under (BPS1) and (BPS2), every realisable partition $\mathcal{P}$ has finitely many cells, and the count is bounded by $N(\mathcal{P}) \leq \mu(\Omega)/\mu_{\min}(\mathcal{P})$, where $\mu_{\min}(\mathcal{P}) = \min_i \mu(c_i) > 0$.*

Proof. By (BPS2) each cell has $\text{diam}(c_i) \geq \delta$, and by (BPS1) $\mu(c_i) > 0$ with the cells disjoint and of finite total measure $\mu(\Omega) < \infty$. If the count were infinite, then $\sum_i \mu(c_i) = \mu(\Omega) < \infty$ would force $\inf_i \mu(c_i) = 0$, contradicting that finitely-described cells carry a uniform positive lower measure bound under finite resolution. Hence the count is finite, and $N(\mathcal{P}) \mu_{\min}(\mathcal{P}) \leq \sum_i \mu(c_i) = \mu(\Omega)$ gives the bound. $\square$

Lemma 2.9 (Recurrence is available). *If (Ω, d, μ) additionally carries a μ -preserving flow $\phi_t: \Omega \rightarrow \Omega$, then for every Borel A with $\mu(A) > 0$ and every $T > 0$ there is $n \geq 1$ with $\mu(A \cap \phi_{nT}^{-1}A) > 0$; μ -a.e. point of A returns to A .*

Proof. The Poincaré recurrence theorem (??): the sets $\{\phi_{-kT}A\}_{k \geq 0}$ all have measure $\mu(A)$; if pairwise disjoint modulo null sets their measures would sum to ∞ , contradicting $\mu(\Omega) < \infty$ from (BPS1). Hence two overlap, giving recurrence. $\square$

Remark 2.10. Theorem 2.9 is used only in ??; the epistemic spine (Section 3 and ????????) uses only the static structure of Axiom 1. We isolate it here to show that the same hypothesis that forces the floor also supports the dynamical realisation.

3 The Boundary–Thickness Theorem

We now prove the central structural fact: a bounded, finitely-resolved knower cannot separate a region from its complement by a set of zero content. Every separator it can realise has positive content, and that content is bounded below by a constant determined by the resolution—the floor.

3.1 Operational separators

The topological boundary ∂A can have measure zero (e.g. a half-space in $\mathbb{R}^m$). The theorem is not about ∂A in the ambient continuum; it is about the separators a finite-resolution knower can actually realise, which must be *unions of cells*.

Definition 3.1 (Realisable region and separator). Fix a partition $\mathcal{P}$ with resolution δ . A region A is $\mathcal{P}$ -realisable if it is a union of cells of $\mathcal{P}$. A $\mathcal{P}$ -separator of a $\mathcal{P}$ -realisable A is the set of cells of $\mathcal{P}$ that meet both A and $\Omega \setminus A$ in the closure—formally,

$$\Sigma_{\mathcal{P}}(A) = \bigcup \{c \in \mathcal{P} : \text{cl}(c) \cap \text{cl}(A) \neq \emptyset \text{ and } \text{cl}(c) \cap \text{cl}(\Omega \setminus A) \neq \emptyset\}.$$

Because a finite-resolution knower can only point to cells, $\Sigma_{\mathcal{P}}(A)$ is the thinnest separator it can name.

Remark 3.2. The shift from ∂A to $\Sigma_{\mathcal{P}}(A)$ is the whole content of “operational.” A continuum knower with unlimited resolution might use ∂A of measure zero; a knower bound by (BPS2) cannot, because it cannot name sub-cell sets. The theorem quantifies the price of that limitation.

3.2 Non-triviality of the separator

Lemma 3.3 (The realisable separator is non-empty). *Let A be $\mathcal{P}$ -realisable with $\mu(A) > 0$ and $\mu(\Omega \setminus A) > 0$, and suppose Ω is connected. Then $\Sigma_{\mathcal{P}}(A) \neq \emptyset$.*

Proof. Suppose $\Sigma_{\mathcal{P}}(A) = \emptyset$. Then no cell meets the closures of both A and $\Omega \setminus A$; hence $\text{cl}(A)$ and $\text{cl}(\Omega \setminus A)$ are disjoint unions of whole cells whose closures do not touch. Writing $U = \text{int}(\text{cl}(A))$ and $V = \text{int}(\text{cl}(\Omega \setminus A))$, the sets U, V are non-empty (each contains a cell of positive measure), open, disjoint, and cover Ω up to a μ -null set; if their closures do not meet, $U \sqcup V$ is a separation of Ω into two non-empty clopen pieces, contradicting connectedness. Hence $\Sigma_{\mathcal{P}}(A) \neq \emptyset$. $\square$

Remark 3.4 (Connectedness). Theorem 3.3 uses connectedness of Ω . If Ω is disconnected, a region can coincide with a component and have empty separator; but then A was not genuinely *divided* from its complement—it was given. The theorem concerns acts of division *within* a connected whole, which is the case of interest (an object carved out of one world). We assume connectedness hereafter and note it where used.

3.3 The thickness lower bound

Definition 3.5 (Boundary thickness). The *boundary thickness* of a $\mathcal{P}$ -realisable region A is the measure of its realisable separator,

$$\beta_{\mathcal{P}}(A) := \mu(\Sigma_{\mathcal{P}}(A)).$$

Theorem 3.6 (Boundary–Thickness Theorem). *Let (Ω, d, μ) be a connected bounded phase space satisfying Axiom 1, and let $\mathcal{P}$ be a realisable partition of resolution δ with minimum cell measure $\mu_{\min} > 0$. Then for every $\mathcal{P}$ -realisable region A with $\mu(A) > 0$ and $\mu(\Omega \setminus A) > 0$,*

$$\boxed{\beta_{\mathcal{P}}(A) \geq \mu_{\min} > 0.}$$

In particular no realisable separator is sharp: $\mu(\Sigma_{\mathcal{P}}(A)) = 0$ is impossible.

Proof. By Theorem 3.3, $\Sigma_{\mathcal{P}}(A)$ contains at least one cell $\mathbf{c}^* \in \mathcal{P}$. By (BPS2) and Theorem 2.8 every cell has $\mu(\mathbf{c}) \geq \mu_{\min} > 0$. Since $\Sigma_{\mathcal{P}}(A)$ is a union of cells and contains $\mathbf{c}^*$,

$$\beta_{\mathcal{P}}(A) = \mu(\Sigma_{\mathcal{P}}(A)) \geq \mu(\mathbf{c}^*) \geq \mu_{\min} > 0.$$

A sharp separator would have $\mu(\Sigma_{\mathcal{P}}(A)) = 0 < \mu_{\min}$, a contradiction. $\square$

Definition 3.7 (Resolution floor). The *resolution floor* of a knower realising partition $\mathcal{P}$ is

$$\beta := \inf_A \beta_{\mathcal{P}}(A) \geq \mu_{\min} > 0,$$

the infimum over all $\mathcal{P}$ -realisable regions of their boundary thickness. By Theorem 3.6 the floor is strictly positive.

Corollary 3.8 (No zero-width cut). *Under Axiom 1, a bounded finite-resolution knower cannot divide a connected whole into a region and its complement by a separator of zero content. Every division it performs deposits content $\geq \beta$ in the separator.*

Proof. Immediate from Theorem 3.6 and ?? $\square$

3.4 Stability and the role of boundedness

The next two results show the bound is robust and that boundedness is essential, not decorative.

Proposition 3.9 (Refinement lowers but never abolishes the floor). *Let $\mathcal{P}' \preceq \mathcal{P}$ be a refinement with $\mu'_{\min} \leq \mu_{\min}$. Then $\beta' \leq \beta$ for the refined floor, and still $\beta' \geq \mu'_{\min} > 0$. The floor decreases monotonically under refinement but is bounded below by the finest realisable cell and never reaches zero at any finite stage.*

Proof. A finer partition can only shrink the separating cells, so $\beta'_{\mathcal{P}'}(A) \leq \beta_{\mathcal{P}}(A)$ for each A , giving $\beta' \leq \beta$. By Theorem 3.6 applied to $\mathcal{P}'$, $\beta' \geq \mu'_{\min} > 0$. By (BPS2) every finite stage carries a positive resolution and hence a positive $\mu'_{\min}$; only in the (non-realisable) limit $\delta \rightarrow 0$ would the bound vanish, and that limit is excluded for any embedded knower. $\square$

Proposition 3.10 (Boundedness is necessary). *The conclusion of Theorem 3.6 fails without (BPS1). On an unbounded space $\mathbb{R}^m$ with Lebesgue measure there exist regions (e.g. half-spaces) whose topological boundary is a hyperplane of $\mathcal{L}$ -measure zero, and one may realise separators of arbitrarily small measure by tapering; moreover, with infinite total measure Theorem 2.8 fails and a partition into cells of vanishing diameter at uniform cost is not excluded.*

Proof. Take $A = \{x_1 \leq 0\} \subset \mathbb{R}^m$. Then $\partial A = \{x_1 = 0\}$ has $\mathcal{L}(\partial A) = 0$, so the topological separator is sharp. The finite-resolution obstruction of Theorem 3.6 relied on Theorem 2.8, whose proof used $\mu(\Omega) < \infty$; with $\mu(\mathbb{R}^m) = \infty$ the measures of infinitely many cells can sum to ∞ without forcing $\inf_i \mu(\mathbf{c}_i) = 0$, so the uniform lower bound $\mu_{\min} > 0$ need not hold. Hence the floor argument does not apply, witnessing the necessity of (BPS1). $\square$

Remark 3.11 (What the theorem does and does not say). Theorem 3.6 does not assert that the ambient continuum has thick boundaries; a half-space in $\mathbb{R}^m$ has a measure-zero topological boundary. It asserts that a *bounded finite-resolution knower*, who can name only unions of positive-measure cells, cannot realise that measure-zero cut: its sharpest nameable separator has measure $\geq \mu_{\min}$. The thickness is a property of the knower-in-the-world, jointly with boundedness, not of the world in the abstract. This is the precise sense in which “items are bounded”: the divisions by which items are individuated are bounded below in content.